Econ 31303 StudyGuide

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Introduction

The begining is the most important part of a journey!

- Syllabus
- Course Calendar

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1 *Symbols, abbr. and formula*

Symbols and abbr

- L *or* N = Labor units, in persons or hours
- K = Amount of physical capital
- H = Amount of human capital
- N = Amount of Natural resources
- MB = Marginal Benefit
- MC = Marginal Cost
- MP = Marginal Product
- HH_s *or* hh_s = house holds
- PPF = Production possibility Frontier
- D = Demand curve
- GDP = Gross Domestic Product
- GNP = Gross National Product
- GNR = Net govt revenue or income
- Y = Consumers' income, National Income, or GDP
- C = consumption
- I = investment
- G = Government purchases
- NX = Net exports, which is (X-M)
- X = value of exports
- M = value of imports
- CPI = Consumer Price Index
- Y/N = Output per person, (Labor) Productivity
- K/N = Capital per worker (per person)
- H/N = Human capital per person
- i = Nominal interest rate
- r = Real interest rate
- π = Inflation or sometimes profit
- P = Price of a good or the price level
- G = government purchases

- $G\&S$ = Goods and services
- $Govt$ = government
- Q = Quantity or amount of the good or service

- S_{pvt} = Private saving
- S_{gov} *or* S_{pub} = Public or government saving
- T = Net taxes, which is (Taxes-govt transfers)
- $RGDP$ = Real GDP
- $NGDP$ = Nominal GDP
- S = Supply Curve, or National Saving (in another context)
- W = Nominal Wage
- w = Real wage
- A = Technology or Total factor productivity
- TFP = Total factor productivity

Formula*

$$Y = C + I + G + NX$$
$$where\ NX = X - M$$

- $GDP = sum\ of\ P \times Q$, for all goods
- $GDP\ deflator = \frac{NGDP}{RGDP} \times 100$
- if $Z = X \cdot Y \Rightarrow \%\Delta Z \approx \%\Delta X + \%\Delta Y$
- if $Z = X/Y \Rightarrow \%\Delta Z \approx \%\Delta X - \%\Delta Y$

$$\begin{aligned} GNP &= GDP + \text{Net factor payments from abroad} \\ &= GDP + \text{Income of domestic factors earned abroad} \\ &\quad - \text{Income of foreign factors earned at home} \end{aligned}$$

- $Y_d = Y + NFP + TR + INT - T$
- $GNR = T - TR - INT$
- $GNR + Y_d = GDP + NFP = GNP$
- $S_{pvt} = (Y + NFP - T + TR + INT) - C$
- $S_{govt} = (T - TR - INT) - G$

Sources of S, national saving:

$$\begin{aligned} S &= Spvt + Sgovt \\ &= (Y + NFP - T + TR + INT - C) + (T - TR - INT - G) \\ &= GNP - C - G \end{aligned}$$

Uses of S:

$$\begin{aligned} S &= GNP - C - G \\ &= (Y + NFP) - C - G \\ &= (C + I + G + NX + NFP) - C - G \\ &= (I + NX + NFP) \\ &= (I + CA) \end{aligned}$$

$Prod.\ function:\ Y = AF(K, N)$

C-D prod. function: $Y = AK^{\alpha}N^{1-\alpha} + u - rate = 100 \times \frac{\#Unemployed}{\#Labor\ Force} + L.F\ partic.\ rate = 100 \times \frac{\#Labor\ Force}{\#Adult\ popn}$

$$\begin{aligned} Cyclical\ unemployment &= u - u^* \\ where, u &= \text{actual unempl} \\ and\ u^* &= \text{Natural of unemployment} \end{aligned}$$

$$and$$

$$\frac{Y - Y^*}{Y^*} = -2(u - u^*)$$

$$and\ \frac{\Delta Y}{Y} = 3 - 2 \cdot \Delta u$$

Chapter 4:

For closed Economy:

$$S = S_{hh} + S_{gov} \tag{1.1}$$
$$= (Y - T - C) + (T - G) \tag{1.2}$$
$$= Y - C - G \tag{1.3}$$

- $Uc \equiv r \cdot P_k + d \cdot P_k = (r + d) \cdot PK$

With tax on capital:

- $MPK^f = \frac{r+d}{1-\tau} P_k$
- $K_{t+1} = K_t + I_t - dK_t \Rightarrow I_t = K_{t+1} - (1 + d)K_t$
- $Tobin's\ q = \frac{V}{P_k \cdot K}$

Equilibrium:

- $AD = AS \Rightarrow \bar{Y} = C(r) + I(r) + G \Rightarrow S = I$

Chapter 7: Asset Markets:

- $\Delta Y/Y = \Delta A/A + \epsilon_K \cdot \Delta K/K + \epsilon_N \cdot \Delta N/N \qquad (2)$
- Steady state: $sf(k) = (n + d)k$
- Golen rule: $MPK = (n + d)$
- $\frac{M_d}{P} = L_d(Y, r + \pi^e)$
- $\Delta M/M - \Delta P/P = \epsilon_Y \Delta Y/Y + \epsilon_i \Delta i/i$
- $V = \frac{P \times Y}{M}$

Chapter 9: IS-LM model:

- $IScurve : S(r; etc) = I(r; etc)$
- $M_S/P = M_d/P \Rightarrow L(r-; Y+, \pi^e-)$

A TBA